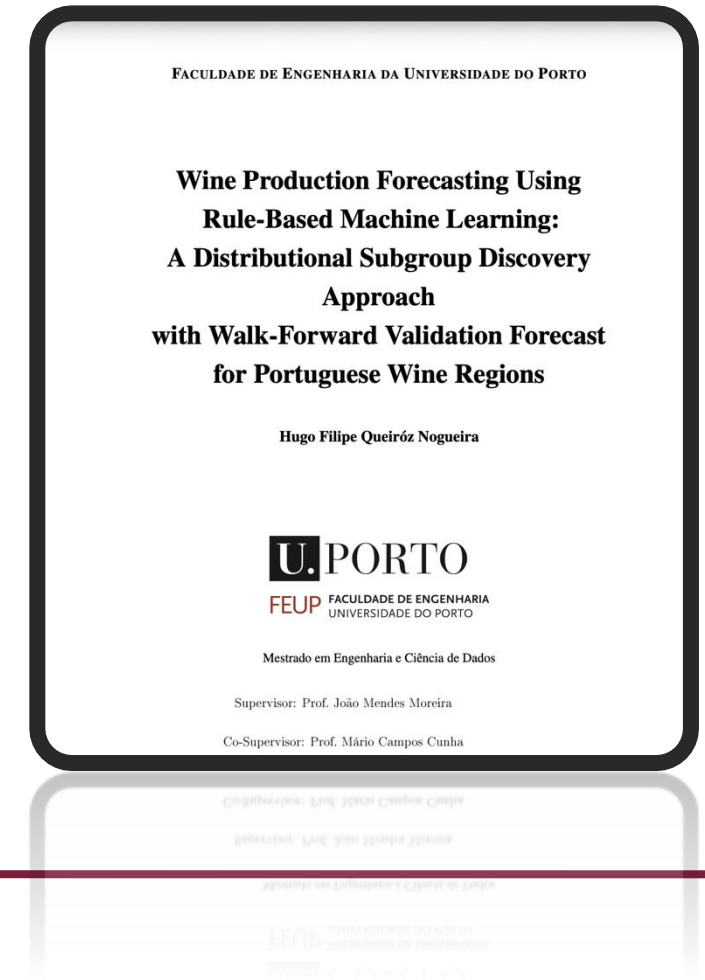


Wine production forecasting with interpretable rules

What can climate explain — and what can it reliably predict?

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Supervisors: João Mendes Moreira · Mário Campos Cunha



The business question is a decision question

Can regional climate and vine development data improve the annual production forecast — early enough to change a decision?

A useful model must add accuracy and preserve trust.

Prediction alone is insufficient when the mechanism cannot be explained or challenged.

Who uses the answer

Producers and
cooperatives
Regulatory bodies
Supply-chain planners

Decisions affected

Harvest planning
Storage and capacity
Risk assessment

RESEARCH PROBLEM

The problem is not only forecasting, but auditable forecasting

THE GAP

Modern ML can predict wine yield, but decision-makers need readable rules they can challenge with domain knowledge.

WHAT A RULE LOOKS LIKE

*IF harvest soil water is moderate
AND post-maturity wet days are low
THEN production is +192 mhl*

NO PRIOR WORK COMBINES ALL FIVE PROPERTIES

Interpretable	Long-term	Predictive	Wine region
Modern ML	no	partial	yes
Subgroup discovery	yes	no	no
This thesis	yes	yes	yes

WHY INTERPRETABILITY MATTERS

Readable by agronomists **without ML expertise**
Each rule carries **support** and a p-value
Can be challenged and domain-validated
Useful for **diagnosis** even when forecasting fails

Why this is hard: the climate signal is real, but diluted

The target combines climate, physiology, human decisions and structural change in one annual regional number.

01

Biology spans years

Yield potential is partly set in the previous season, before the harvest year begins.

02

Aggregation hides variation

Thousands of vineyards differ in soil, altitude, practices and local weather.

03

Structure changes over time

Policy, planted area, producers and market conditions shift the production baseline.

04

The sample is small

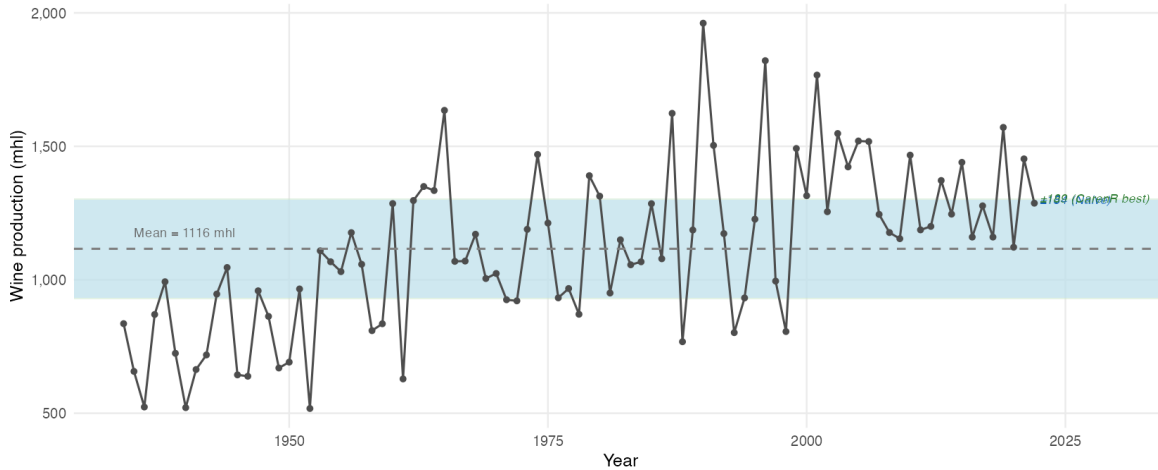
Only 80–89 annual observations must support 59 engineered features.

NATURAL EXPERIMENT

The same pipeline meets two different histories

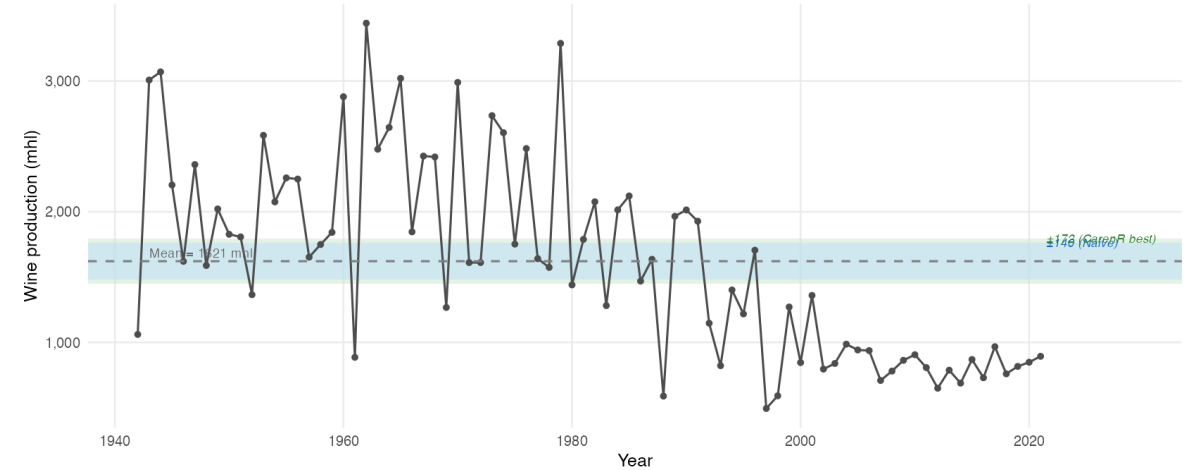
Wine Production (Douro (RDD), 1934–2022) with MAE Error Bands

Mean = 1116 mhl, SD = 307 mhl, CV = 27%



Wine Production (Vinho Verde (RVV), 1942–2021) with MAE Error Bands

Mean = 1621 mhl, SD = 755 mhl, CV = 47%



DOURO · 89 years

Gradual +80% structural rise; CV 27%

VINHO VERDE · 80 years

Sharp structural decline; CV 47%

GRAPEVINE CYCLE

Current production begins in the previous growing season

YEAR y0 · POTENTIAL

**Primordia
formation**

**Reserve
building**

Temperature and light set the upper bound on bunch number.

Prior water and thermal stress shape carbohydrate reserves.

**CARRY-
OVER**

YEAR y1 · REALISATION

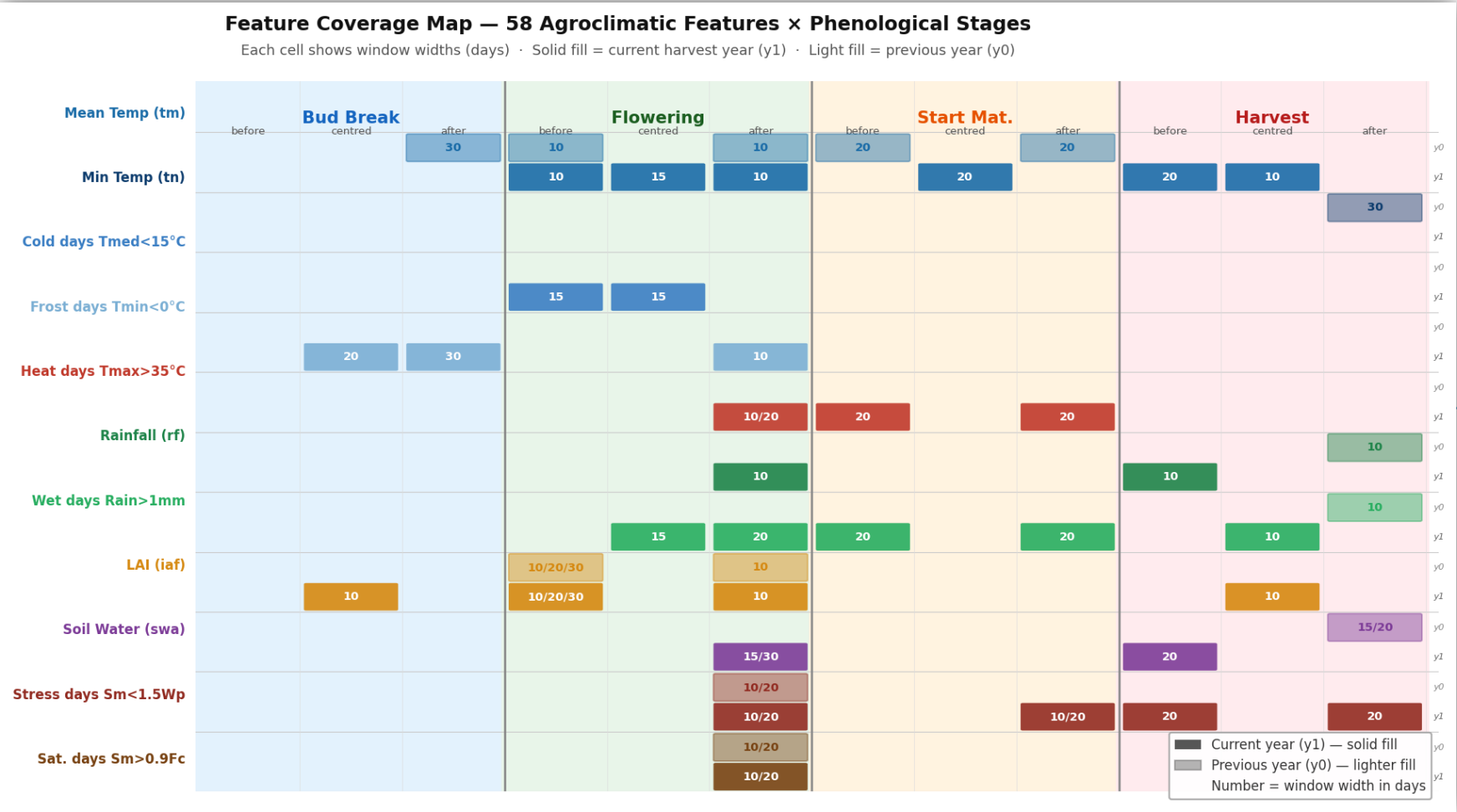
Bud break

**Flowering &
fruit set**

**Veraison to
harvest**

Current-year temperature, rain, canopy and soil water determine fruit set and berry development.

Features follow the biology, not the calendar



year	Wine_mhl	tm_fl_y1_c15	tm_fl_y1_b10	tm_fl_y1_a10	tm_fl_y0_b10	tm_fl_y0
1934	835.6	[18.3 22.4]	[18.1 21.5]	[18.8 24.3]	[16 18]	[18.8 24.
1935	656.7	[13.8 16.8]	[14.3 16]	[17 18.8]	[18 21.5]	[18.8 24.
1936	523.3	[16.8 18.3]	[16 18.1]	[12.3 17]	[14.3 16]	[17 18.8]
1937	870.0	[13.8 16.8]	[14.3 16]	[12.3 17]	[16 18]	[12.3 17]
1938	992.6	[13.8 16.8]	[14.3 16]	[12.3 17]	[14.3 16]	[12.3 17]
1939	724.5	[16.8 18.3]	[14.3 16]	[18.8 24.3]	[14.3 16]	[12.3 17]

32,842 / 29,902
daily records

-> 59 annual
predictors

-> 4 phenological
anchors

-> current + previous
year

The research gap sits between explanation and prediction

Modern machine learning

Handles nonlinear interactions
Often strong predictive accuracy
Sometimes decision logic remains opaque

**Can we
have
both?**

Rule-based discovery

Produces explicit IF–THEN rules
Auditable support and significance
Predictive value is unproven here

No prior work systematically evaluated distributional subgroup discovery in walk-forward wine production forecasting.

Two goals separate historical insight from future utility

GOAL 1

DISCOVER

Which agroclimatic conditions distinguish above- and below-trend production years?

Discovery

RQ1 · *Can CarenR's KS-test-based distributional rule induction extract statistically validated, non-trivial patterns from non-stationary agricultural time series?*

Validation

RQ2 · *Are the discovered rules genuine within-era climate signal, or are they era-membership artefacts introduced by imperfect linear detrending?*

GOAL 2

VALIDATE

Can rule-based models beat the historical median in genuine future periods?

Prediction

RQ3 · *Can rule-based models (CarenR, RIPPER, M5Rules) outperform a Naive Median baseline in walk-forward prediction of detrended production residuals?*

Diagnosis

RQ4 · *What structural conditions (era composition, detrend quality, signal strength) determine when rule-based prediction succeeds or fails?*

A controlled comparison isolates the role of data structure

SAME

Feature engineering
Detrending logic
Rule learners
Evaluation metric

DIFFERENT

Climate regime
Structural trajectory
Noise level
Era composition

2
regions

89 + 80
annual observations

THE CONTRAST TURNS TWO CASE STUDIES INTO A NATURAL EXPERIMENT.

CarenR is the primary lens among five model families

PRIMARY METHOD

CarenR

Distributional rules + prediction variants

- 1 **Discretise** Climate features become equal-frequency bins.
- 2 **Search** Level-wise search (Apriori-style) combines feature-bin conditions.
- 3 **Compare** KS test: subgroup residuals vs global residuals.
- 4 **Retain** Keep significant rules with sufficient support.

OUTPUT · IF climate conditions → shifted production-residual distribution

COMPARATORS

RIPPER

Rule learner + prediction comparator

M5Rules

Rule learner + prediction comparator

Decision Tree

Prediction-only non-rule ML comparator

Naive baselines

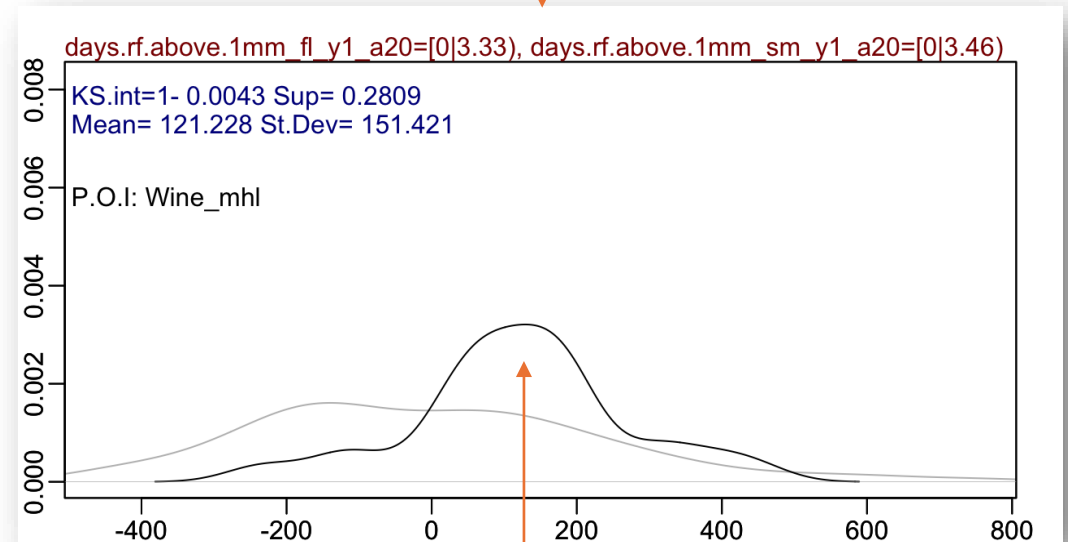
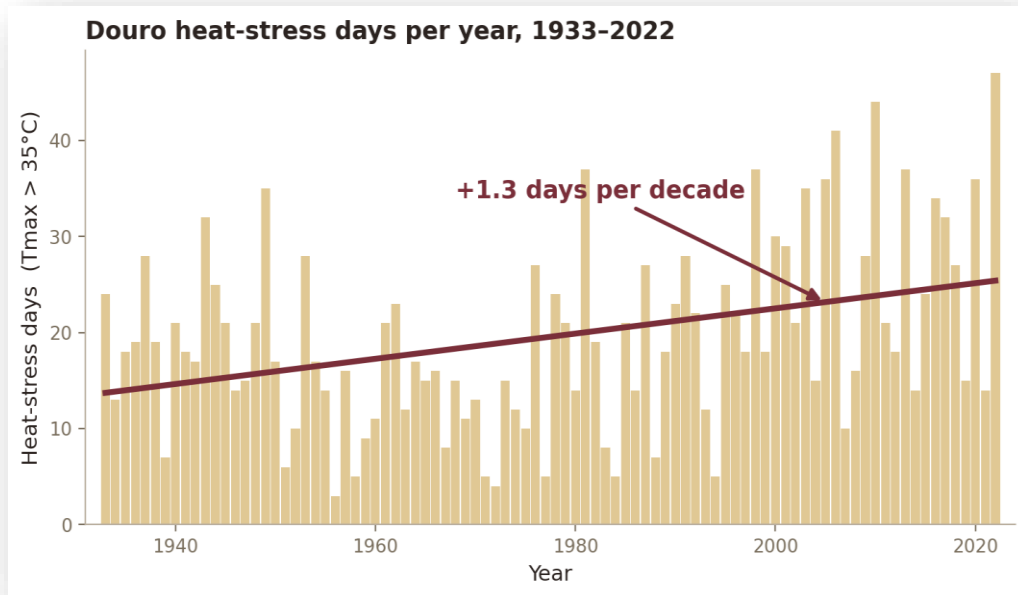
Prediction-only references

Roles: CarenR links rules + prediction; RIPPER/M5Rules cross-check rules; Decision Tree and naive baselines test prediction only.

Goal 1 finds statistically exceptional subgroups

CarenR asks a distributional question

DO YEARS MATCHING THESE CLIMATE CONDITIONS HAVE A PRODUCTION-RESIDUAL DISTRIBUTION THAT DIFFERS FROM ALL YEARS?



A RULE YOU CAN AUDIT

IF dry post-flowering **AND** moderate harvest soil water

THEN +121 mhl above trend

support 28% · $p = 0.004$

Goal 1 succeeded: compact rules reveal regional drivers

DOURO

48 rules

Harvest soil water appears in 54% of rules.

Dry post-flowering conditions and bounded heat repeatedly associate with above-trend production.

VINHO VERDE

20 rules

Heat-stress days appear in up to 50% of rules.

Harvest soil water (25%) and cool, moist conditions dominate; the LAI family appears in 4 of 20 rules.

The same variable can have a different direction across regions — interpretation must remain context-specific.

The strongest CarenR rules point to water and canopy structure

DOURO · top 5 CarenR rules

4 of 5 are above-trend: dry flowering and bounded heat define good years.

- | | | |
|----------|--|-----------------|
| 1 | Dry flowering + dry maturity | +121 mhl |
| | 28.1% support · p=0.0043 | |
| 2 | High LAI + wet flowering | -131 mhl |
| | 20.2% support · p=0.0084 | |
| 3 | Dry + moderate heat after flowering | +113 mhl |
| | 25.8% support · p=0.0089 | |
| 4 | Harvest water + carry-over dryness | +192 mhl |
| | 21.3% support · p=0.0095 | |
| 5 | Dry flowering + bounded heat | +115 mhl |
| | 24.7% support · p=0.0110 | |

VINHO VERDE · top 5 CarenR rules

4 of 5 are below-trend: cool, moist conditions dominate the Atlantic region.

- | | | |
|----------|---|-----------------|
| 1 | Moist harvest + no heat stress | -338 mhl |
| | 21.2% support · p=0.0198 | |
| 2 | Cool flowering + prior soil saturation | -268 mhl |
| | 21.2% support · p=0.0216 | |
| 3 | Moist post-flowering + low heat | -365 mhl |
| | 27.5% support · p=0.0294 | |
| 4 | Warm prior budburst + mild flowering | +311 mhl |
| | 25.0% support · p=0.0298 | |
| 5 | Cool post-flowering + wet carry-over | -292 mhl |
| | 23.8% support · p=0.0299 | |

Cross-algorithm finding

Across methods: **harvest soil water and LAI/canopy survive.**

CarenR, RIPPER and M5Rules all select them, so the signal is not tied to one optimization criterion.

Exact CarenR rule output: intervals behind the top rules

Each row is an IF condition set; Delta is the median production residual shift from trend.

DOURO (RDD) · top 5 exact rules

Mostly above-trend: dry post-flowering / maturity windows and bounded heat.

#	Exact conditions from CarenR output	support / p	Delta
1	Wet days post-Flowering [0-3.3 d] AND Wet days post-Maturity [0-3.5 d]	28.1% ~25 yr p=0.004	+121
2	LAI at Flowering [0.74-0.88 m2/m2] AND Soil water post-Flowering [390-404 mm]	20.2% ~18 yr p=0.008	-131
3	Wet days post-Flowering [0-3.3 d] AND Wet days post-Maturity [0-3.5 d] AND Heat days post-Flowering [0-0.6 d]	25.8% ~23 yr p=0.0089	+113
4	Soil water pre-Harvest [74-101 mm] AND Wet days post-Harvest prev.yr [0-2.3 d] AND Wet days post-Maturity [0-3.5 d]	21.3% ~19 yr p=0.009	+192
5	Wet days post-Flowering [0-3.3 d] AND Heat days pre-Maturity [0-3.7 d]	24.7% ~22 yr p=0.0110	+115

VINHO VERDE (RVV) · top 5 exact rules

Mostly below-trend: cool, moist conditions dominate the Atlantic region.

#	Exact conditions from CarenR output	support / p	Delta
1	Soil water pre-Harvest [61-97 mm] AND Heat days pre-Maturity [0-1.9 d] AND Heat days post-Maturity [0-2.4 d] AND Heat days post-Flowering [0-1.1 d]	21.2% ~17 yr p=0.0198	-338
2	Temp at Flowering [13.8-17.1 deg C] AND Temp post-Flowering [12.6-17.5 deg C] AND Wet-soil days post-Flowering prev.yr [17.5-20 d]	21.2% ~17 yr p=0.0216	-268
3	Soil water post-Flowering [143-234 mm] AND Heat days post-Maturity [0-2.4 d]	27.5% ~22 yr p=0.0294	-365
4	Temp post-Budburst prev.yr [11.2-13.5 deg C] AND Cold days at Flowering [0-3.4 d] AND Heat days post-Flowering [0-1.1 d]	25.0% ~20 yr p=0.0298	+311
5	Temp post-Flowering [12.6-17.5 deg C] AND Heat days post-Maturity [0-2.4 d] AND Wet-soil days post-Flowering prev.yr [17.5-20 d]	23.8% ~19 yr p=0.0299	-292

Three optimization criteria recover the same feature families

Feature	CarenR	RIPPER	M5Rules	Confidence
Harvest soil water	54% of RDD rules 25% of RVV rules	RVV Rule 3	-2.3/mm RDD -8.1/mm RVV	HIGH universal
LAI / canopy family	4 of 20 RVV rules minor RDD signal	RVV Rules 1-2	-2064/unit RVV variable selected	HIGH universal
Previous-year summer temperature	RVV carry-over	RVV Rule 3	+17.2/C RVV	HIGH RVV focus
Wet days after flowering	21% of RDD rules	not selected	not selected	MODERATE CarenR-specific

Agreement across three independent methods is the strongest evidence that the rule features reflect real data structure, not algorithmic noise.

Goal 1 - Two checks confirm the discovered signal

01

AGGRESSIVE DETRENDING

RDD signals persist after LOESS; in RVV, wet-days persist while LAI and soil water are era-linked.

→ The signal is real, but not equally stable across regions.

02

INDEPENDENT ALGORITHMS

RIPPER and *M5Rules* converge on harvest soil water, LAI and previous-year temperature.

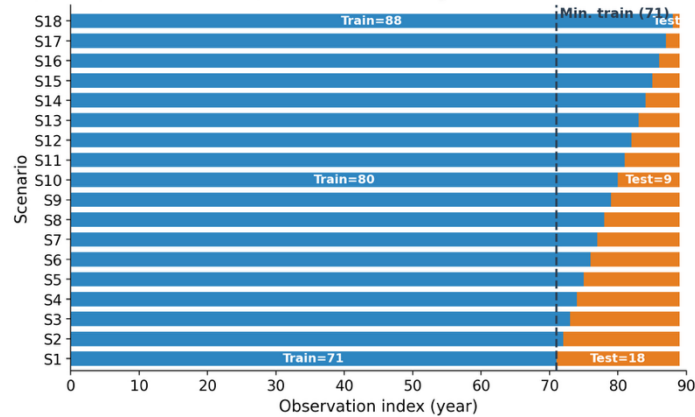
→ The signal is not CarenR-specific.

Discovery is credible — but credibility is not yet predictive utility.

Goal 2 recreates prediction-time information

Walk-Forward Expanding-Window Validation Protocol

RDD (Douro) — 18 scenarios, minimum training window = 71 observations (80%)



Expanding-window validation

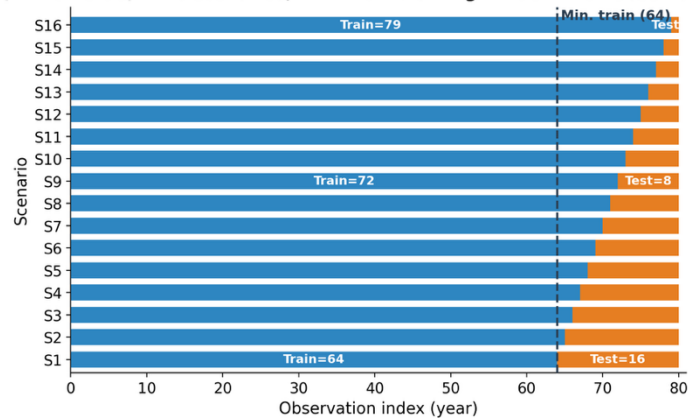
18 scenarios · Douro

16 scenarios · Vinho Verde

All preprocessing stays inside each training fold:

- detrending
- discretisation
- feature selection
- rule induction

RVV (Vinho Verde) — 16 scenarios, minimum training window = 64 observations (80%)



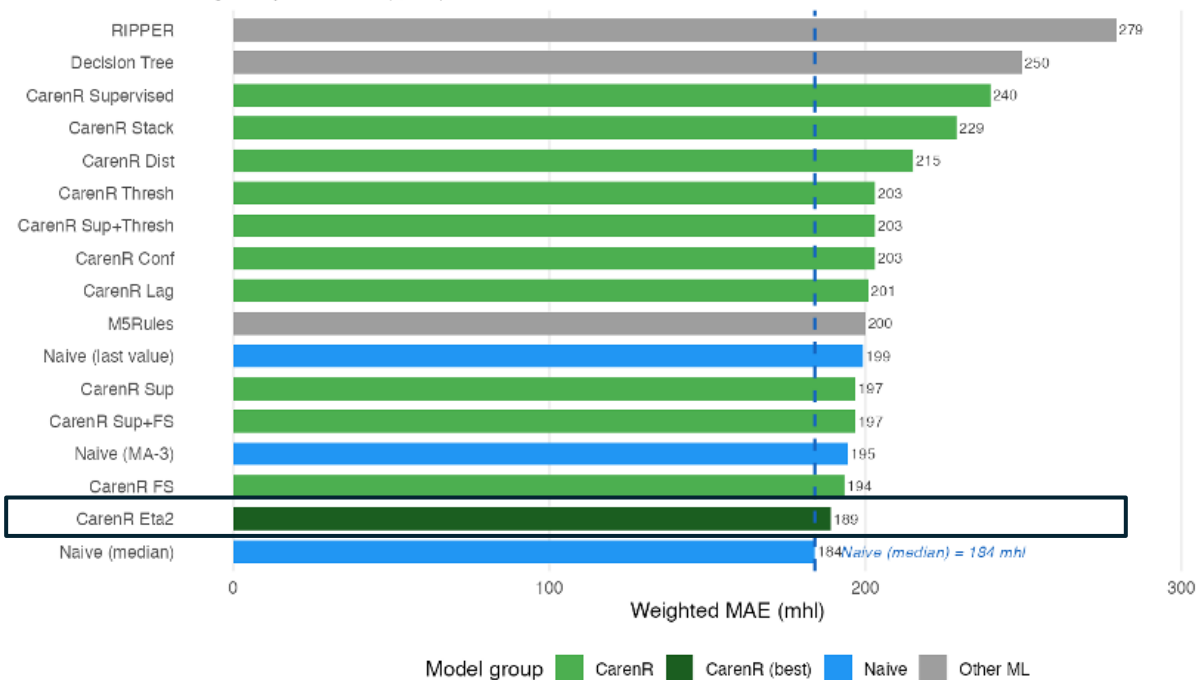
Training window (expanding) Test window (shrinking)

17 models · weighted MAE · lower is better

No rule-based model beats the median overall

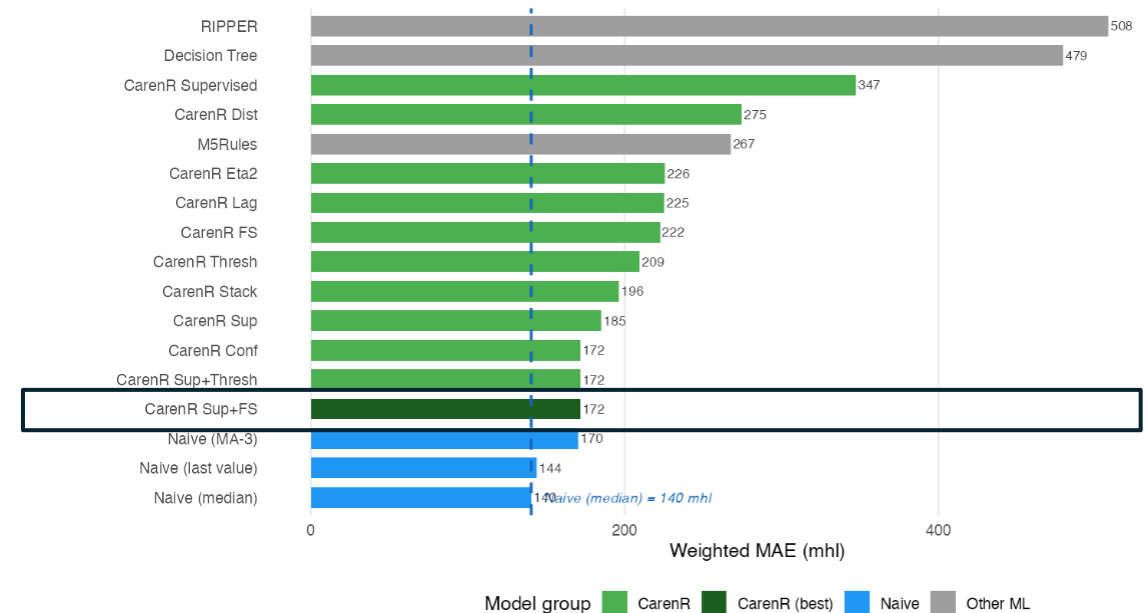
Model Comparison: Weighted MAE across 18 Walk-Forward Scenarios (RDD)

Weighted by test-set size (n_test) — lower is better



Model Comparison: Weighted MAE across 16 Walk-Forward Scenarios (RVV)

Weighted by test-set size (n_test) — lower is better



RDD · 189 vs 184 mhl

3% worse · $p = 0.468$

RVV · 172 vs 140 mhl

22% worse · $p = 0.004^*$

*Among the 9 scenarios where RVV rules fired, they significantly harmed prediction.

A real association can still be too weak to forecast

Pattern discovery and forecasting answer different questions.

DISCOVERY

**IS THE SUBGROUP
DISTRIBUTION DIFFERENT?**

With $|r| \approx 0.47$, the association is detectable across 80–89 historical years.

≠

PREDICTION

**DOES THE RULE REDUCE
FUTURE ERROR?**

Climate explains $\approx 22\%$ of residual variance; $\approx 78\%$ remains unobserved or noisy.

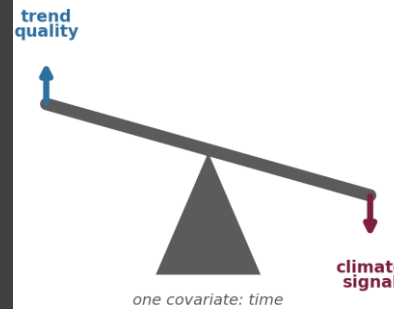
Better detrending can remove the signal rules need

DECISION 1

MODEL THE STRUCTURAL TREND

A better trend leaves smaller test residuals and lowers the naive error floor.

TENSION



DECISION 2

EXTRACT CLIMATE SIGNAL

Stable rules need residual structure that is large enough to detect and exploit.

RVV LOESS: baseline improves 140 → 90 mhl, but rule induction falls to zero.

Time alone cannot cleanly separate industry structure from climate.

Five scientific contributions clarify why rules work and fail

1 Two-decision framework separates trend quality from residual climate-signal extraction.

2 · Walk-forward evaluation

17 models, 18/16 scenarios,
and all preprocessing inside
each fold.

3 · Sensitivity analysis

Eleven CarenR variants show
the best feature-selection
strategy is problem-specific.

4 · Era-composition analysis

Identifies when rule-based
prediction may become
viable, as hypothesis for
future work.

**5 · CROSS-ALGORITHM VALIDATION: RIPPER AND M5RULES INDEPENDENTLY CONFIRM
CAREN R FEATURE FAMILIES.**

The next evidence should separate climate from structure

LIMITATIONS

- Time is the only detrending covariate, so structural change and climate remain partly mixed.
- The walk-forward test has only 18 and 16 scenarios; the 5/5 RDD era-balanced result is exploratory.
- Regional annual aggregates hide sub-regional weather and vineyard heterogeneity.

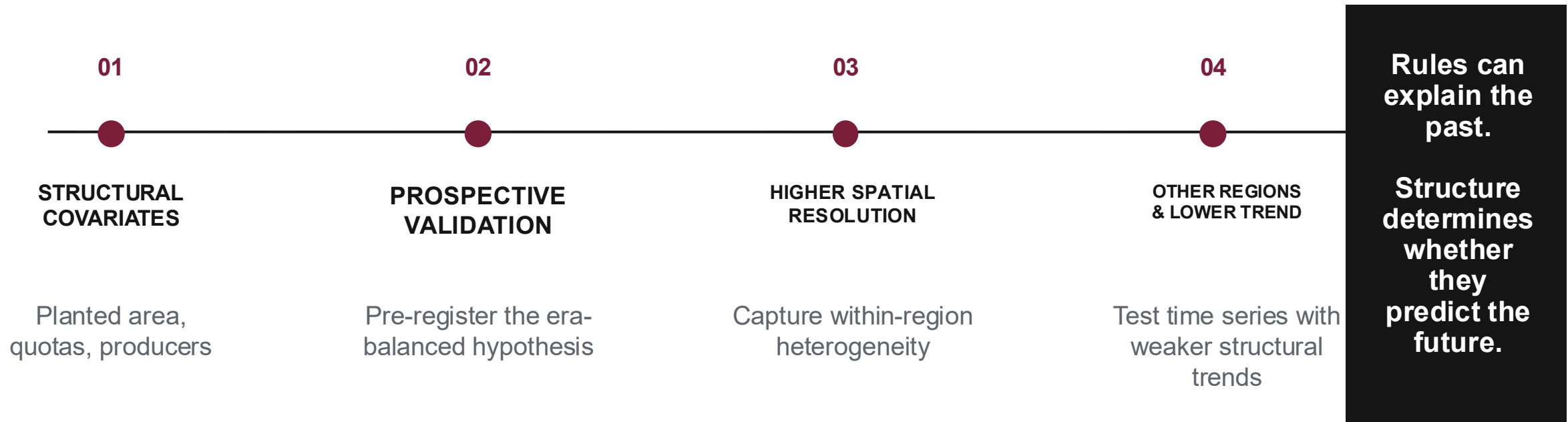
FUTURE WORK

- Add structural covariates: planted area, quotas and certified producers.
- Pre-register and prospectively validate the era-balanced hypothesis.
- Test other regional time series with weaker trends and use higher-resolution climate data.

THE THESIS DOES NOT CLAIM AN OPERATIONAL FORECAST YET;
IT SHOWS WHEN INTERPRETABLE RULES ARE MEANINGFUL, WHEN THEY FAIL, AND WHAT EVIDENCE IS
NEEDED NEXT

NEXT STEPS

The next step is to model structure explicitly



Thank you

Questions